

Tim Csernica

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Education

California Institute of Technology

Pasadena, CA

PhD in Chemistry

June 2024

- Advisor: John Eiler
- Committee: Geoff Blake, Brian Stoltz, & Rudy Marcus
- Thesis: Exploring Nature's Fingerprints with Isotopic Distributions

University of Chicago

Chicago, IL

B.S. with Honors in Chemistry

June 2018

- Advisor: Steven Sibener
- Phi Beta Kappa

Professional Experience

2025-Present	Postdoctoral Scholar , Civil and Environmental Engineering, Northwestern University
2024-2025	Staff Scientist (Remote) , Geological and Planetary Sciences, Caltech
2024	Postdoctoral Scholar , Geological and Planetary Sciences, Caltech
2018-2024	Graduate Researcher , Chemistry and Chemical Engineering, Caltech
2017	Undergraduate Researcher , Chemical Engineering, Technion – Israel Institute of Tech.
2015-2018	Undergraduate Researcher , Chemistry, University of Chicago
2015	University of Chicago Mathematics REU , Mathematics, University of Chicago

Grants and Awards

2024	Merkin Innovation Seed Grant: Natural Abundance Stable Isotopes of Metabolites for Cancer Diagnosis (non-PI)	\$100,000
2021	Distinguished Alumnus: Lewisburg Area High School	-
2017	Norman H. Nachtrieb Memorial Award for Undergraduate Studies in Chemistry	\$1,500

Peer-Reviewed Publications

- [1] **Csernica, T.**, Eiler, J. M., and Sessions, A. L. "[Site-specific, Multi-Isotope Characterization of DDT Isolated from Sediment Samples.](#)" *Marine Pollution Bulletin*. **2026**.
- [2] Wang, X., Liu, C., Han, J., Liu, J., **Csernica, T.**, Zhang, N., Ding, H., Gropp, J., McIntosh, J., Li, S., Xu, S., & Ellam, R. M. "[Clumped isotopes tracing methane cycle and budget: a review.](#)" *Earth Science Reviews*. **2026**.
- [3] **Csernica, T.**, Moran, J. J., Fraga, C. G., and Eiler, J. M. "[Simultaneous observation of 2H and 13C enrichment of methyl phosphonic acid via Orbitrap-IRMS with applications to nerve agent forensics.](#)" *Talanta*. **2025**.

- [4] Wang, X., Biying, C., Chen, L., Dong, G., **Csernica, T.**, Zhang, N., Liu, J., Shuai, Y., Liu, C., Xu, Z., Li, S., and Xu, S. "[Biogenic methane clumped isotope signatures: insights from microbially enhanced coal bed methane and machine learning.](#)" *Fuels*. **2024**.
- [5] Ammanamanchi, N., Yester, J., Bargaje, A. P., Thomas, D., Little, K. C., Janzef, S., Francis, K., Weinberg, J., Johnson, J., Seery, T., Harris, T. H., Funari, B. J., Rose-Felker, K., Zinn, M., Miller, S. A., West, S. C., Feingold, B., Zhou, H., Steinhauser, M. L., **Csernica, T.**, Michener, R., and Kühn, B. "[Elimination of ¹⁵N-thymidine after oral administration in human infants.](#)" *PLOS One*. **2024**.
- [6] **Csernica, T.**, Sessions, A. L., and Eiler, J. M. "[High-Dimensional Isotomics Part 2: Observations of Over 100 Constraints on Methionine's Isotome.](#)" *Chemical Geology*. **2023**.
- [7] Weiss, G. M., Sessions, A. L., Julien, M., **Csernica, T.**, Yamada, K., Gilbert, A., Freeman, K. H., and Eiler, J. M. "[Intramolecular carbon isotope variation of labeled alanine.](#)" *International Journal of Mass Spectrometry*. **2023**.
- [8] **Csernica, T.**, Bhattacharjee, S., and Eiler, J. M. "[Accuracy and precision of ESI-Orbitrap-IRMS observations of hours to tens of hours via reservoir injection.](#)" *International Journal of Mass Spectrometry*. **2023**.
- [9] **Csernica, T.**, and Eiler, J. M. "[High-Dimensional Isotomics Part 1: A Mathematical Framework for Isotomics.](#)" *Chemical Geology*. **2023**.

Other Publications

- [1] **Csernica, T.**, "[Extinction in single and multi-type branching processes.](#)" For University of Chicago Mathematics REU, **2015**.

Presentations

Invited Oral Presentations

- [1] **Inaugural International Orbitrap-IRMS Users Meeting.** Goldschmidt Conference. **2025**. *Declined*
- [2] **ThermoFisher Orbitrap Workshop,** American Geophysical Union Conference. "Data Quality Tests for Orbitrap-IRMS Data, Including Lessons from the 100 Isotopologue Challenge." **2023**.
- [3] **ThermoFisher Orbitrap Workshop,** Orbitrap Isotope Users Meeting. "Expanding the Orbitrap-IRMS Frontier." **2023**.
- [4] **Invited Talk,** American Society for Mass Spectrometry. "Mapping the Isotope Anatomies of Organic Molecules at Natural Abundance & Beyond". **2022**.
- [5] **Invited Talk,** Yuk Yung Lunch Seminar. "Primitive Genetics and the Origin of Life." **2020**.

Oral Presentations

- [1] **American Chemical Society.** "Orbitrap Mass Spectrometry for Stable Isotope Forensics: A Case Study of Sarin Precursors." **2024.**
- [2] **International Symposium on Isotopologues.** "High-Dimensional Isotomics: Observation and Interpretation of Over 100 Isotopic Constraints on Methionine." **2022.**

Poster Presentations

- [1] **Goldschmidt Virtual.** "Reconstructing Extraterrestrial Organic Synthesis via Site-Specific Isotopic Measurements. **2021.**
- [2] **Simons Collaboration on the Origins of Life Annual Meeting, Virtual.** "Isotopic Characterization of Nucleobases from a Possible Extraterrestrial Reaction Network. **2020.**
- [3] **Gordon Conference on the Origin of Life.** "Site-specific Isotopic Fingerprinting of an Abiotically Plausible Adenine Synthesis. **2020.**

Teaching & Mentoring Experience

2026	Mentor, Center for Synthetic Biology REU	Northwestern
2019	Chemical Synthesis and Characterization for Chemical Engineering, TA	Caltech
2018-2019	General Chemistry, TA	Caltech

Teaching Workshops

<i>Northwestern University Practicum: The Science of How Students Learn</i>	Spring 2026
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